

Lab: Neural-Inspired World Models for Robots

Objective

Design robotic cognitive architectures inspired by biological neural systems. You will compare hippocampal spatial representations with SLAM algorithms, analyze how the brain builds and updates world models, and design a neural-inspired cognitive architecture for a mobile robot.

Prerequisites

- Completion of the corresponding module.md lecture material
- Familiarity with Active Inference concepts (generative models, free energy, prediction errors)
- Basic pseudocode and diagram skills

Part 1: Hippocampal Navigation Models

The hippocampus supports spatial navigation through place cells, grid cells, and head direction cells. Design a robotic equivalent:

1. How do place cells (firing at specific locations) map to a robotic localization system?
2. How do grid cells (periodic spatial firing patterns) map to metric position encoding?
3. How do head direction cells map to compass/heading estimation?
4. How does hippocampal replay (replaying past trajectories during rest) relate to planning in Active Inference?

Write your response here

Part 2: Cortical Hierarchy for Object Recognition

Design a hierarchical object recognition system inspired by the ventral visual stream (V1 -> V2 -> V4 -> IT cortex):

1. Specify what each level represents (edges, textures, parts, whole objects).
2. Define top-down generative predictions at each level.
3. Define bottom-up prediction errors at each level.
4. How does this compare to a standard convolutional neural network? What are the key differences from an Active Inference perspective?

Write your response here

Part 3: Memory Systems for Robots

Biological cognition relies on multiple memory systems. Design robotic equivalents:

1. **Working memory** (prefrontal cortex): short-term task state, current goals.
2. **Episodic memory** (hippocampus): specific past experiences and locations.
3. **Semantic memory** (temporal cortex): general knowledge about objects and categories.
4. **Procedural memory** (basal ganglia, cerebellum): learned motor skills.

How do these memory systems interact in a hierarchical generative model?

Write your response here

Part 4: Cognitive Architecture Comparison

Compare three cognitive architectures:

Feature	Brain-Inspired	Classical AI (SLAM + planner)	Active Inference
Spatial representation			
Memory organization			
Learning mechanism			
Planning strategy			

Write your response here

Summary Table

Concept	Classical Robotics	Active Inference	Your Design
Core mechanism			
Uncertainty handling			
Adaptation			
Key advantage			

References

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- Parr, T., Pezzulo, G., & Friston, K. J. (2022). *Active Inference*. MIT Press.
- Pio-Lopez, L., Nizard, A., Friston, K., & Pezzulo, G. (2016). Active Inference and robot control. *Journal of the Royal Society Interface*, 13(122).